

Automated Cancer Type Classification from Pathology Reports **Using Transformer-Based NLP**

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Data Pipeline & Preprocessing

01

Download & Merge :

9,522 TCGA reports from GitHub, merged with cancer type labels using patient ID (first 12 chars of filename). 32 unique cancer types after merge.

02

Text Cleaning :

Lowercase -> remove non-letter chars -> strip English stop words -> drop tokens ≤ 2 chars. Reduces 80,000 raw tokens to 15,000 meaningful medical terms per report.

03

TF-IDF Vectorization :

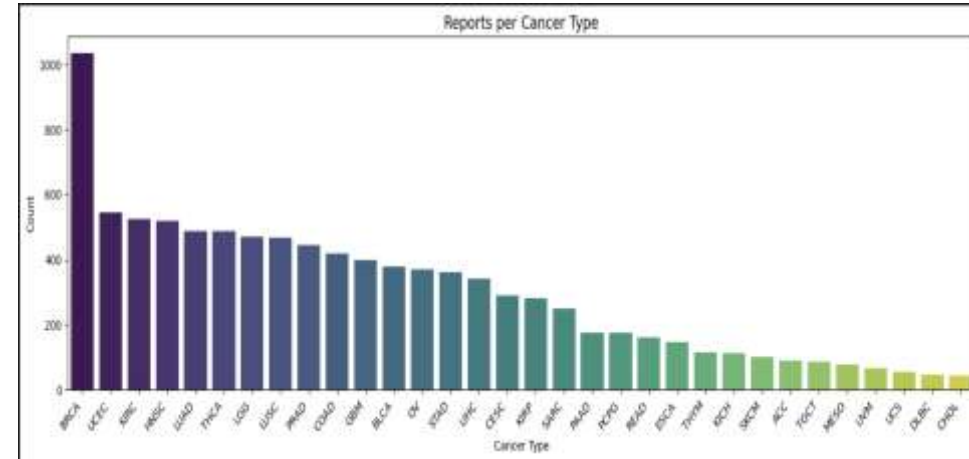
15,000 features, unigram + bigram, sublinear TF scaling. Bigrams capture 'lung adenocarcinoma' as one feature. fit on train, transform on test.

04

Stratified 80/20 Split :

7,617 train / 1,905 test. Stratified so all 32 cancer types appear proportionally.
Critical: BRCA has 1,034 samples; CHOL has only 43.

Dataset Distribution - Top 8 Cancer Types



BRCA (1,034) is 24× more common than CHOL (43). Stratified splitting ensures all 32 types appear in train and test sets.

Sample cancer types in dataset :

BRCA (Breast) · LUAD (Lung Adeno) · KIRC (Kidney Clear Cell) · PRAD (Prostate) · UCEC (Uterine) · GBM (Brain) · OV (Ovarian) · THCA (Thyroid) · STAD (Stomach) · COAD (Colon) · READ (Rectal) · BLCA (Bladder) ... 20 more types

Four Models, Two Approaches

❖ Shared transformer fine-tuning config:

5 epochs · LR 1e-5 · Batch 16 · Warmup 15% · Weight decay 0.02 · Early stopping patience=3 · max length=256 · eval per epoch · best checkpoint auto-saved

Random Forest (Classical) :

- 200 trees, majority vote on TF-IDF features
- Robust to high-dimensional sparse data
- CPU-only trains in under 5 minutes

95.3%

accuracy

Linear SVM (Classical) :

- Optimal hyperplane across 32 classes
- C=0.5 tuned · maxiter=5000
- Fastest model , no GPU needed

97.7%

accuracy

Google BERT (Transformer) :

- Pre-trained: Wikipedia + Book Corpus
- 12 layers · 110M parameters · bidirectional
- Fine-tuned [CLS] token for classification

95.7%

accuracy

PubMedBERT (Transformer) :

- Pre-trained: 14M PubMed biomedical papers
- Natively understands medical vocabulary
- Lower eval loss than BERT from epoch 1

97.1%

accuracy

Implementation & Evaluation - Classical Models

95.3%

Random Forest

accuracy

97.7%

Linear SVM

accuracy

97.6%

SVM F1 Score

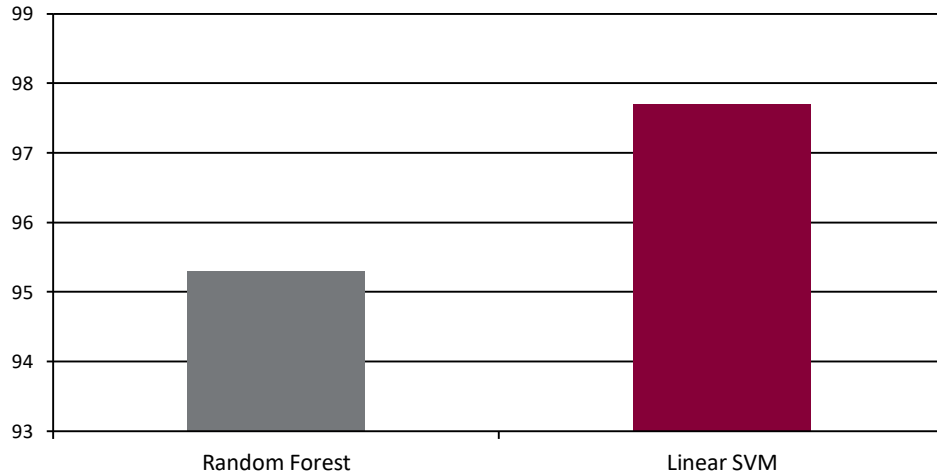
weighted F1

32

Cancer Types

all classified

Accuracy - Classical Models (%)



What stood out :

Linear SVM at 97.7% (strongest model overall).

Beat every model including the deep learning ones. TF-IDF creates a very high-dimensional sparse feature space and SVM finds the optimal hyperplane in exactly that space. Consistent clinical language in pathology reports plays directly to SVM's strengths.

Per-class precision highlights (SVM):

BRCA: 0.99 · PRAD: 0.99 · THCA: 1.00 · Most types: F1 > 0.95

Random Forest at 95.3% (solid ensemble baseline).

200 trees on CPU in under 5 minutes. Stratified split ensured CHOL (43 samples) appeared in both train and test sets.

Both classical models trained without any GPU.

F1 = harmonic mean of precision & recall. Weighted F1 97.6% means the model is precise AND catches most true cases across all 32 types.

Implementation & Evaluation — Transformer Models

95.7%

Google BERT

accuracy

97.1%

PubMedBERT

accuracy

97.0%

PubMedBERT F1

weighted F1

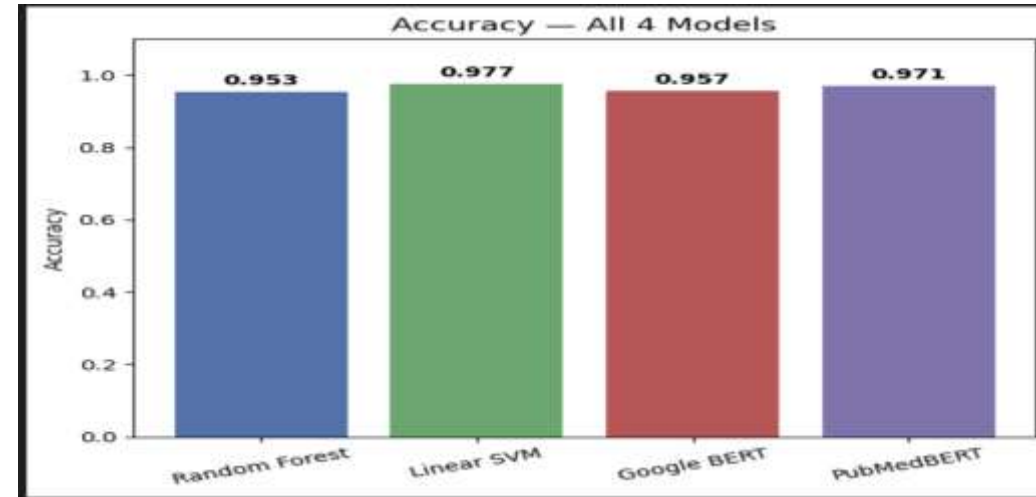
0.127

Final Eval Loss

PubMedBERT

PubMedBERT - 5-Epoch Training Progress (* = best checkpoint)

Epoch	Train Loss	Val Loss	Accuracy	F1 (weighted)
1	0.3176	0.2047	96.17%	95.88%
2	0.1500	0.1166	97.59%	97.57%
3 *	0.0698	0.1056	97.90%	97.89%
4	0.0421	0.1190	97.10%	97.02%
5	0.0287	0.1270	97.10%	97.02%



❖ Why PubMedBERT > Google BERT:

PubMedBERT was pre-trained on 14 million PubMed biomedical articles. Words like adenocarcinoma, biopsy, metastasis, and resection are native vocabulary tokens. General BERT had to learn these from scratch during fine-tuning and domain-specific pre-training makes a measurable difference.

Error Analysis - What Are Confusion Matrices?

What is a Confusion Matrix?

A grid where rows = true cancer type and columns = predicted cancer type. Numbers on the diagonal are correct predictions. Off-diagonal = mistakes.

True label (rows)	BRCA	LUAD	KIRC
BRCA	207	0	1
LUAD	0	97	2
KIRC	2	3	105

- SVM misclassifies only 2.3% of 1,905 test samples about 44 reports total. A perfect model has zeros everywhere except the diagonal.
- *Most errors occur between biologically similar cancers: KIRC<->KIRP (both kidney), COAD<->READ (colon vs rectal), LUAD<->LUSC (lung subtypes). Even expert pathologists sometimes disagree on these pairs.*

What Our Results Revealed :

❖ Strong diagonal all 4 models

All models show concentrated numbers on the diagonal. High accuracy is real across all 32 cancer types.

❖ Errors are biologically meaningful

KIRC<->KIRP (kidney) · COAD<->READ (colon/rectal) · LUAD<->LUSC (lung). These are clinically similar errors are structured, not random noise.

❖ PubMedBERT: fewest off-diagonal errors

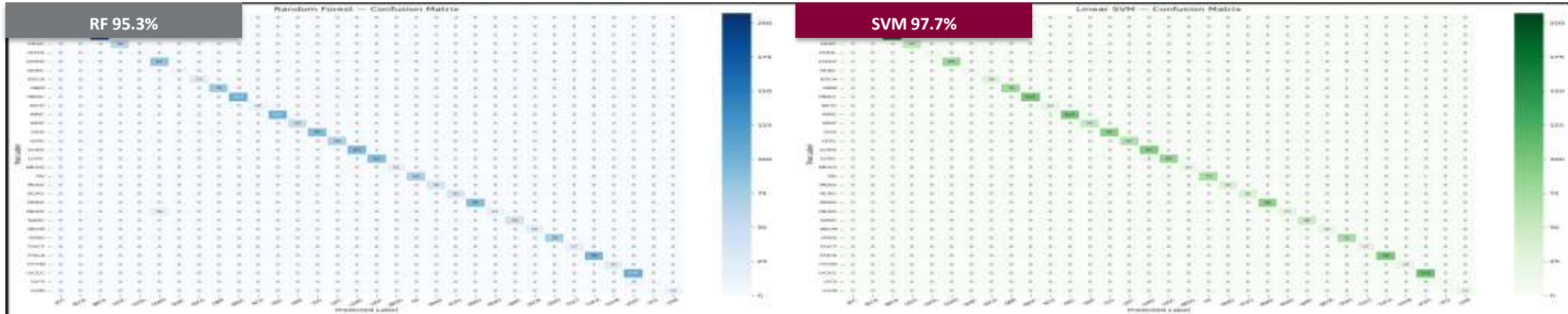
Cleanest matrix among transformers, consistent with its lower eval loss (0.127 vs 0.200) and biomedical pre-training.

❖ Clinical takeaway

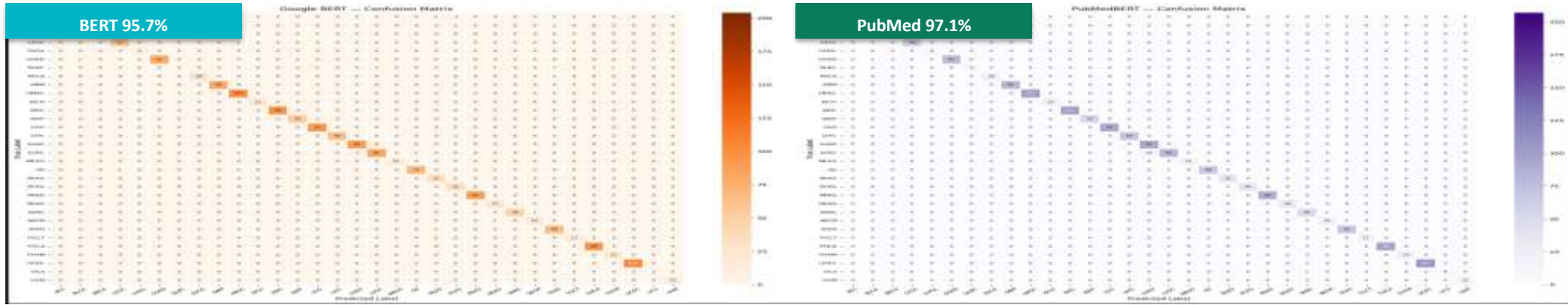
The model's errors mirror real human diagnostic challenges. This is structured uncertainty not model failure.

Error Analysis - Confusion Matrices (All 4 Models)

Classical: Random Forest (blue) | Linear SVM (green)



Transformer: Google BERT (orange) | PubMedBERT (purple)



Strong diagonal in all 4 models. Errors cluster between biologically similar types (KIRC<->KIRP, COAD<->READ, LUAD<->LUSC) - expected, not random.

Conclusion & Future Work

Model	Accuracy	F1	Notes
Random Forest	95.3%	95.0%	Baseline - CPU only, < 5 min
Linear SVM	97.7%	97.6%	Best overall - no GPU needed
Google BERT	95.7%	95.4%	General transformer model
PubMedBERT	97.1%	97.0%	Best domain-specific model

Key Takeaways :

- NLP achieves 95 - 98% accuracy classifying 32 cancer types from raw text.
- Linear SVM + TF-IDF is competitive with deep learning and far cheaper to deploy.
- Domain pre-training matters: PubMedBERT beats general BERT on medical text.
- 97.7% accuracy could provide pathologists with an automated second opinion, reducing turnaround from days to seconds.

What I Would Explore Next :

Address Class Imbalance

SMOTE or class-weighted loss for rare types like CHOL (43 samples)

Ensemble SVM + PubMedBERT

Combining both best models could push accuracy past 99%

Sub-type Classification

Predict tumor grade, stage, or molecular subtype from same reports

Clinical API Deployment

Lightweight REST endpoint for real-time pathology support

THANK YOU!